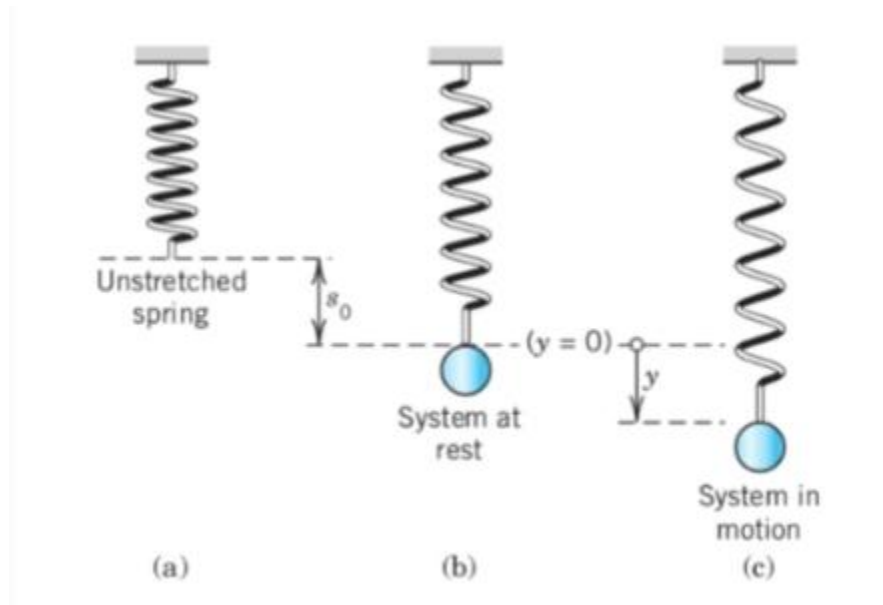


Project Report

Ayesha Imran 09884, Widad Khan 09758

Spring Mass System and Second Order ODEs

Introduction:



Spring mass systems are among the most fundamental models in engineering and applied mathematics. They describe how mechanical systems respond to external forces and provide the foundation for analyzing vibrations, oscillations, and dynamic stability in structures, vehicles, aircraft, and control systems. A spring mass system is modeled by a second-order ordinary differential equation because the system's behavior is determined by the relationship between mass, acceleration (the second derivative of displacement), and the forces acting on it, like the spring's restoring force.

The motion of a mass attached to a spring is governed by Newton's Second Law and leads naturally to a second-order linear ordinary differential equation. This project explores:

1. The mathematical principles behind such systems.
2. Derivation of the spring mass ODE.
3. Analytical and computational solution techniques.
4. Interpretation of the resulting motion in real world notations.

Core Mathematical Principles behind spring mass systems:

1.1 Newton's Second Law:

The law states that the net force acting on a body is equal to the product of its mass and its acceleration.

$$F_{\text{net}} = m (d^2y / dt^2)$$

In a spring mass system, the displacement $y(t)$, velocity $y'(t)$, and acceleration $y''(t)$ of the mass must be related to the forces acting on it. Newton's Law establishes this relationship directly by asserting that acceleration being represented by the second derivative of displacement is produced by the sum of all forces acting on the mass.

1.2 Hooke's Law (Spring Force):

This law states that the force F required to extend or compress a spring is directly proportional to its displacement y from its equilibrium position. The formula is:

$$F = -ky$$

Where k is the spring constant, which is the measure of the spring's stiffness and the negative sign indicates force towards equilibrium. The law applies within an object's elastic limit, meaning the object returns to its original shape after the force is removed.

1.3 Damping Force:

A damping force in a mass spring system is a force that resists motion, dissipating energy and causing oscillations to die out over time; the energy loss occurs due to friction and air resistance. This force is proportional to the velocity of the mass, the formula is:

$$F_d = -c (dy / dt)$$

where $c > 0$ is the damping coefficient. Without damping, oscillations in an ideal mass spring system would continue forever, but the damping force, reduces the amplitude and duration of these vibrations.

1.4 External Forces:

The forces that influence the motion in a spring mass system are the spring force, damping force and any external forces for example impulse, constant load, harmonic force $F_0 \sin(\omega t)$ and periodic input. Any external force applied to a mass spring system from outside can causes it to deviate from its natural motion. When a periodic external force is applied, the system can exhibit driven oscillations and under certain conditions resonance takes place where the system begins to vibrate at maximum amplitude.

2 Derivation of the spring mass ODE:

We make the following assumptions:

1. The spring must obey hooke's law.
2. Damping is proportional to velocity (viscous damping meaning the faster the object moves the stronger the damping force becomes)
3. Motion is one dimensional; for this case restricted to the y-axis (the spring only moves along one line and all the forces act along that same direction)
4. No nonlinearities meaning the springs stiffness and damping does not change as the mass moves. Nothing in the system behaves differently at different displacements or speeds.

2.1 Ideal spring mass system (No damping):

To find the net force we will add all the forces acting on the mass.

1. Spring restoring force (hooke's law): $F_x = -ky(t)$; where k is the spring constant, large value of k refers to a stiffer spring.
2. Damping force: $F_d = -cy'(t)$; where c is the damping coefficient, c measures the strength of resistance.
3. We assume no frictional or external forces so, $F_{\text{ext}}(t)=0$.
4. Net force in y-direction = $ma = my''(t)$

$$F_{\text{net}} = F_s + F_d + F_{\text{ext}}(t)$$

After substitution:

$$my''(t) = -ky(t) - cy'(t) + 0$$

After rearranging:

$$my''(t) + cy'(t) + ky(t) = 0$$

This is how we derive the second order linear ODE for a no damping spring mass system.

3 Analytical and computational solution techniques:

3.1 Auxiliary Equation:

$my'' + cy' + ky = 0$; this is a homogenous ODE since no term completely depends on t .

We assume $y(t) = e^{rt}$, where $y(t)$ is the displacement of the mass at time t , and we substitute this in the ODE:

$y' = re^{rt}$, $y'' = r^2e^{rt}$, after substitution we get:

$mr^2e^{rt} + cre^{rt} + ke^{rt} = 0$, we will divide by e^{rt} and we know that $e^{rt} \neq 0$. We get:

$$mr^2 + cr + k = 0$$

now we will solve the quadratic:

$$r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

Discriminant:

$$\Delta = c^2 - 4mk$$

The nature of the discriminant gives us 3 cases of damping.

3.2 Case 1 (Underdamped):

An underdamped spring mass system is one where damping (like friction or air resistance) is present but weak, causing the system to oscillate back and forth around its equilibrium point with a gradually decreasing amplitude until it eventually settles; one where the oscillations “die out” over time.

$$c^2 - 4mk < 0 \text{ or } c^2 < 4mk$$

This means we have complex roots.

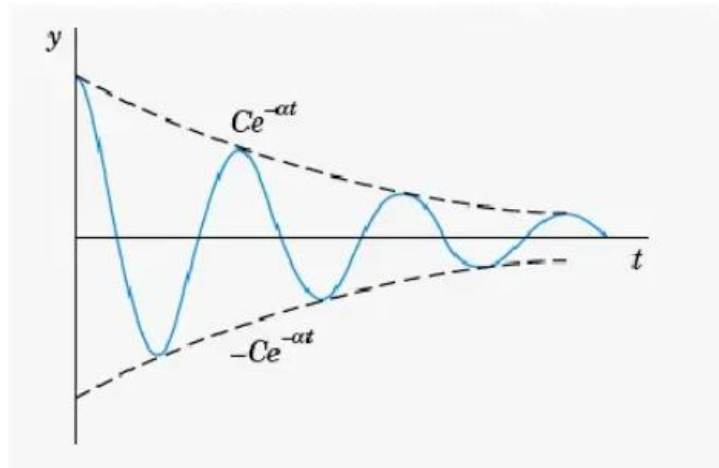
$$r = -\frac{c}{2m} \pm i\omega$$

Solution:

$$y(t) = e^{-\frac{c}{2m}t} (A \cos(\omega t) + B \sin(\omega t)), \text{ where A and B are the initial conditions.}$$

A is the initial position $y(0)$.

B is the initial velocity $y'(0)$.



3.3 Case 2 (critically damped):

A critically damped spring mass system returns to its equilibrium position as fast as possible without oscillating.

$$c^2 - 4mk = 0 \text{ or } c^2 = 4mk$$

This means we have real repeated roots.

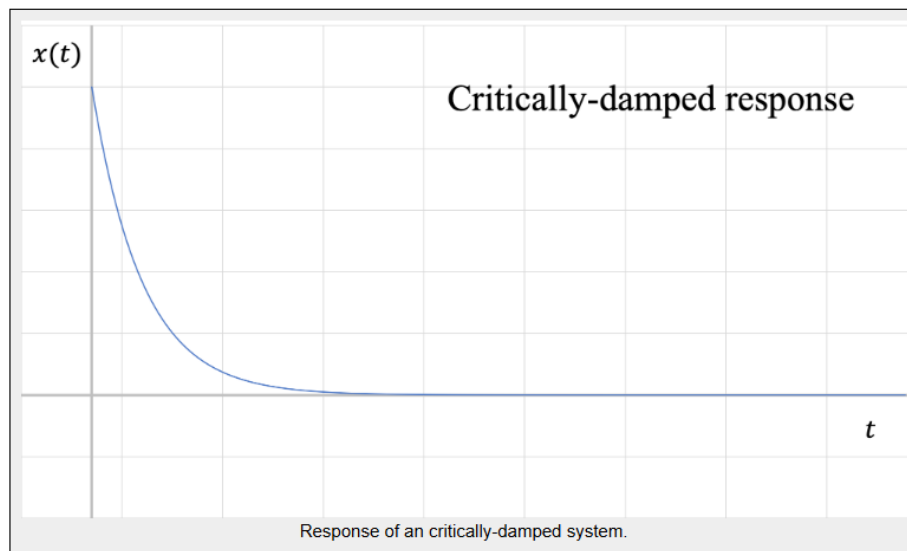
$$r = -\frac{c}{2m}$$

Solution:

$$y(t) = (A + Bt)e^{-\frac{c}{2m}t}, \text{ where } A \text{ and } B \text{ are the initial conditions.}$$

A is the initial position $y(0)$.

B is the initial velocity $y'(0)$.



3.4 Case 3 (overdamped):

An overdamped spring mass system is one where damping (like friction or air resistance) is so strong that the mass returns slowly to its equilibrium position without oscillating.

$$c^2 - 4mk > 0 \text{ or } c^2 > 4mk$$

This means we have two distinct real roots.

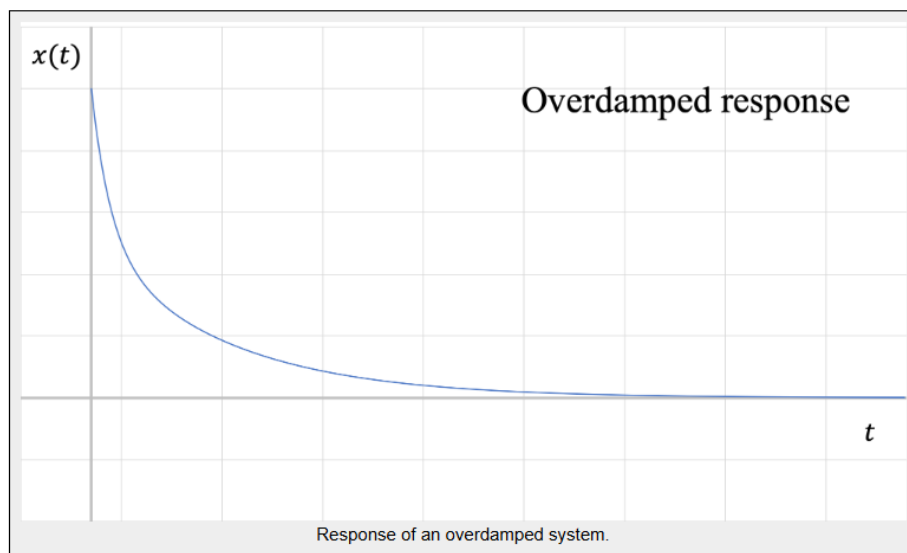
$$r_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}, \text{ where } r_1 \neq r_2$$

Solution:

$$y(t) = Ae^{r_1 t} + Be^{r_2 t}, \text{ where } A \text{ and } B \text{ are the initial conditions.}$$

A is the initial position $y(0)$.

B is the initial velocity $y'(0)$.



Summary of Cases I–III

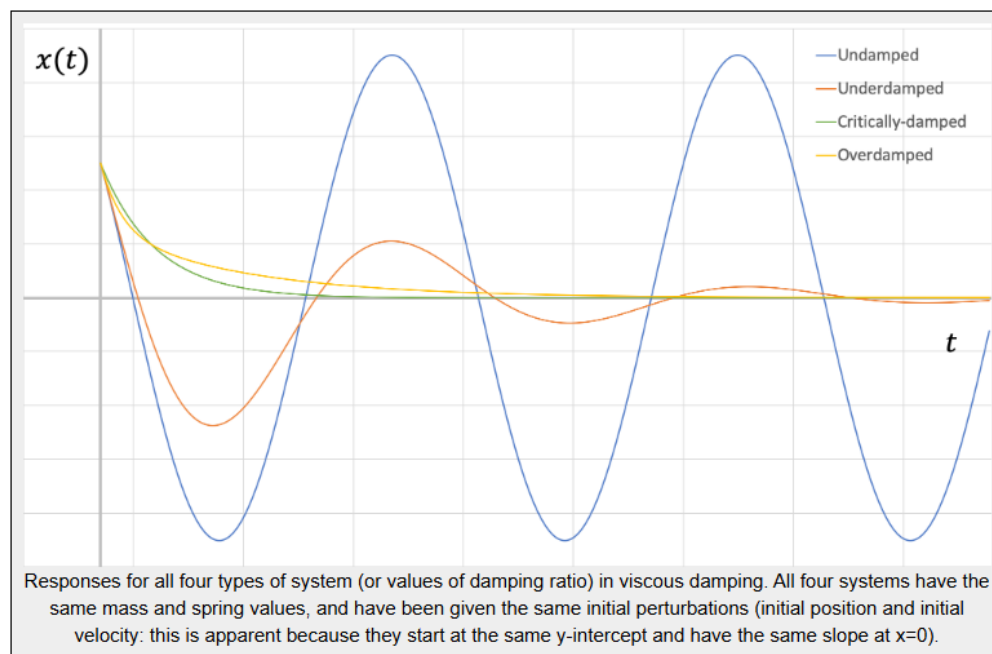
Case	Roots of (2)	Basis of (1)	General Solution of (1)
I	Distinct real λ_1, λ_2	$e^{\lambda_1 x}, e^{\lambda_2 x}$	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
II	Real double root $\lambda = -\frac{1}{2}a$	$e^{-ax/2}, xe^{-ax/2}$	$y = (c_1 + c_2 x)e^{-ax/2}$
III	Complex conjugate $\lambda_1 = -\frac{1}{2}a + i\omega$, $\lambda_2 = -\frac{1}{2}a - i\omega$	$e^{-ax/2} \cos \omega x$ $e^{-ax/2} \sin \omega x$	$y = e^{-ax/2}(A \cos \omega x + B \sin \omega x)$

4 Interpretation of the resulting motion in real world notations:

1. An example of underdamped systems is car suspension without shock absorbers; the car keeps bouncing for a long time after a bump because there is very little damping.
2. Critically damped systems are used in professional shock absorbers in vehicles; they reduce the bouncing of the car after a bump because there is very little damping.
3. Overdamped systems are used in very stiff machinery mounts; they are designed to resist deformation, so they prevent oscillations by absorbing or restricting the machines motions.

Conclusion:

The spring mass system is a foundational model in engineering because it captures essential dynamic behavior using a simple second order ODE. By adjusting mass, damping, and stiffness, we can predict and control oscillatory motion, ensuring stability in mechanical and structural systems. Through analytical techniques, we derived clear solutions for the three types of damping. These solutions reveal how real systems respond to external forces and dissipate energy. This framework extends to more advanced engineering applications such as vibration control, seismic design, and dynamic system optimization.



References:

1. <https://en.wikipedia.org/wiki/Damping?>
2. https://mechanicsmap.psu.edu/websites/16_one_dof_vibrations/16-2_viscoous_damped_free/16-2_viscoous_damped_free.html?
3. <https://faculty.washington.edu/seattle/physics227/reading/reading-3b.pdf?>
Pages 4 & 5
4. https://math.libretexts.org/Courses/Cosumnes_River_College/Math_420%3A_Differential_Equations_%28Breitenbach%29/06%3A_Applications_of_Linear_Second_Order_Equations/6.02%3A_Spring-Mass_Problems_%28With_Damping%29?
5. Sir Asghar Ali's lecture 15 notes
6. Engineering Mathematics: Modeling With 2 Order ODE: Mass-Spring System:
<https://www.scribd.com/presentation/463651852/Lecture-5?v=0.372>